INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Joshna Amirthaa Mariappan

Role: Documentation and Presentation
Project Title: Flight Delay Analysis and Visualization

WEEK 6 – Feature Engineering and Methodology Documentation

During the sixth week, I focused on documenting the feature engineering activities and updating the methodology section of the project report. Feature engineering was an important part of the analysis because it involved preparing and organizing relevant variables to make the dataset more useful for further investigation.

I coordinated with the team to understand the features selected for analysis and how they contributed to studying flight delay patterns. I documented the purpose of feature preparation and its connection to the overall project workflow.

I also reviewed the methodology followed by the team, ensuring that the documentation explained the different stages in the

correct order. These stages included data collection, data quality assessment, preprocessing, exploratory data analysis, feature preparation, statistical analysis, and visualization.

Along with this, I worked on improving the technical explanations in the report. I focused on explaining why feature engineering was required and how selecting relevant variables could help make the analysis more meaningful.

I also began organizing the information required for the statistical analysis section. Since correlation analysis and Principal Component Analysis were planned as part of the project, I prepared the report structure to accommodate their methodology, results, and interpretations.

Another task during this week was coordinating with the visualization team to understand how the prepared variables would eventually be represented through charts and dashboard components.

Outcome and Learning:

By the end of this week, I had improved the methodology and feature engineering documentation. I gained a better understanding of how different stages of data preparation connect to statistical analysis and final visualization.